

Digital Therapeutics with Virtual Reality and Sensors

Filipa Ferreira-Brito

LASIGE, Faculdade de Ciências,
Universidade de Lisboa, Portugal
and ISAMB, Faculdade de Medicina,
Universidade de Lisboa, Portugal
filipa.brito@campus.ul.pt

João Ferreira

LASIGE, Faculdade de Ciências,
Universidade de Lisboa, Portugal
jpferreira@lasige.di.fc.ul.pt

Madalena Vieira

LASIGE, Faculdade de Ciências,
Universidade de Lisboa, Portugal
fc53346@alunos.fc.ul.pt

João Guerreiro

LASIGE, Faculdade de Ciências,
Universidade de Lisboa, Portugal
jpguerreiro@fc.ul.pt

Tiago Guerreiro

LASIGE, Faculdade de Ciências,
Universidade de Lisboa, Portugal
tjvg@di.fc.ul.pt

ABSTRACT

Digital Therapeutics (DTx) is an emerging field within Digital Health that is rapidly growing and holds promise for addressing anxiety-related disorders. Moreover, the combination of virtual reality (VR) and wearable sensors holds great potential for developing affordable, accessible, and effective DTx solutions. In this paper, we present our conception of a fully customised DTx for managing and supporting anxiety-related disorders therapy. We highlight five key assumptions that should be considered when developing such a DTx. These assumptions include the customisation of virtual environments (VEs); integration of physiological data and patient feedback; the use of dashboards for data visualisation; session replay and identification of peak response moments; and patient empowerment throughout the therapeutic process. By embracing these assumptions, we describe the main components constituting this DTx and emphasise fundamental features to fulfil the identified assumptions. Furthermore, we explore some of the potential benefits and challenges of DTx applied to mental health.

KEYWORDS

DTx, Virtual reality, Sensors, Exposure therapy, Anxiety-related disorders, Mental health

ACM Reference Format:

Filipa Ferreira-Brito, João Ferreira, Madalena Vieira, João Guerreiro, and Tiago Guerreiro. 2023. Digital Therapeutics with Virtual Reality and Sensors. In *Adjunct Proceedings of the 2023 ACM International Joint Conference on Pervasive and Ubiquitous Computing & the 2023 ACM International Symposium on Wearable Computing (UbiComp/ISWC '23 Adjunct)*, October 08–12, 2023, Cancun, Quintana Roo, Mexico. ACM, New York, NY, USA, 4 pages. <https://doi.org/10.1145/3594739.3611327>

1 INTRODUCTION

One of the strategic objectives defined in the World Health Organization's (WHO) Global Strategy on Digital Health 2020-2025 [4] is to

enhance access to effective and affordable digital healthcare services by fostering the development of people-centred health systems empowered by digital health. The term "digital health" encompasses a wide range of platforms that utilise digital technology as a valuable tool to address various challenges in the healthcare field, including medical care, nursing services, and healthcare support [11]. One emerging area within digital health is digital therapeutics (DTx).

Digital therapeutics are innovative therapeutic interventions utilising evidence-based digital technologies driven by high-quality software programs, targeting disease treatment, prevention, monitoring, and management [11]. Moreover, DTx also aims to empower patients by creating digital platforms that enable patients to participate in their own treatment decisions [3]. Despite being in its early stages, DTx had a global value estimated at 1.8 billion USD in 2018 and is expected to reach 8.9 billion USD in 2027 (cited by [3, 12]). This makes DTx one of the most promising areas of research and development in the coming years [3]. Scientific evidence has demonstrated the efficacy and cost-effectiveness of DTx in various healthcare fields, including smoking cessation [9], and substance abuse [10]. Moreover, a recent report has identified chronic obstructive pulmonary disorder (COPD), diabetes, weight loss, developmental disorders, and psychological disorders, including depression and anxiety-related disorders, as some of the most promising areas for developing DTx [3].

Virtual Reality (VR) has been increasingly applied in DTx, particularly DTx platforms for exposure therapy in anxiety-related disorders [3]. Exposure therapy (ET), involves patients learning to cope with problematic cognitions and behaviours by gradually and systematically exposing them to feared stimuli [8]. Exposure therapy is considered one of the most successful cognitive-behavioural therapy (CBT) approaches for anxiety-related disorders [8]. In this context, VR allows for the provision of tailored and realistic scenarios for patients to be exposed to feared stimuli without compromising their safety, confidentiality, or anonymity [5].

The access to objective data on a patient's emotional state using sensors collection data on skin conductance response, heart-rate variability, blood volume pulse, and eye tracking has also been identified as a promising source of reliable information regarding the efficacy of virtual reality exposure therapy (VRET).



This work is licensed under a Creative Commons Attribution-NonCommercial International 4.0 License.

UbiComp/ISWC '23 Adjunct, October 08–12, 2023, Cancun, Quintana Roo, Mexico
© 2023 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-0200-6/23/10.
<https://doi.org/10.1145/3594739.3611327>

This article presents our vision for DTx in the context of anxiety-related disorders, specifically combining personalised virtual environments (VEs) with sensors. In our conception of DTx for anxiety-related disorders, we propose a platform that enables the optimisation of VRET by fulfilling the following goals:

- **Customisation of VEs:** the platform should allow for the customisation of VEs based on individual patient needs before and during VRET sessions. This flexibility ensures the therapy is tailored to each patient's requirements and promotes personalised treatment.
- **Integration of physiological data and patient feedback:** the platform should enable the recording and synchronising of sensor readings (i.e., physiological data) with patients' comments during VRET sessions. By combining subjective and objective data, the DTx platform provides a more comprehensive understanding of the patient's responses and a more holistic assessment of the therapeutic process.
- **Dashboards for data visualisation:** the platform should visually represent the data collected (VEs scenarios and physiological data) over multiple VRET sessions. This visualisation enables clinicians to track patients' clinical progress and identify trigger stimuli, situations, or critical moments during sessions.
- **Session replay and identification of peak response moments:** the platform should allow for the reproduction of therapy sessions and enable clinicians to pin/identify specific time windows where patients exhibited heightened behavioural or enhanced physiological responses to the VEs. This feature allows clinicians and patients to review and analyse specific moments of emotional arousal, facilitating a deeper understanding of triggers and potential areas for intervention.
- **Patient empowerment:** by providing a user-friendly interface for data visualisation, the platform is a tool for promoting meaningful clinician-patient discussions and ensuring personalised treatment plans. This fosters a comprehensive understanding of the therapeutic journey, encourages active patient involvement in determining the next steps of the therapeutic process, and might enhance adherence to therapeutic objectives.

2 DTX FOR VRET IN ANXIETY-RELATED DISORDERS

After conducting an initial literature review on VR's current practices, challenges and opportunities for ET [5], our research team has identified three potential scenarios for inclusion in a DTx solution for anxiety-related disorders. These scenarios include arachnophobia, public-speaking fear, and fear of heights. Upon identification of these scenarios, our research team focused on two key elements of the DTx platform: VEs and informative dashboards. Subsequent meetings were held to explore and discuss what features should be included in each component to ensure the fulfilment of the five previously identified goals.

2.1 Customisation of VEs

Customisation of VEs plays a fundamental role in our approach to DTx for VRET in anxiety-related disorders. Figure 1 shows some of the current scenarios created, and Figure 2 shows the control interface of the fear of public-speaking scenario. The ability to customise VEs before and during VRET sessions enables clinicians to synchronise the therapeutic process and tasks according to individual patient needs and clinical progress.

This flexibility allows clinicians to effectively prepare and adapt the VEs to meet patients' expectations and needs. Ultimately, this approach can potentially optimise performance, enhance patient therapeutic adherence, and increase the overall efficacy of VRET. In Table 1, we have outlined key features that have been identified (up to this point) as significant for the customization of VEs in the three anxiety-related disorders mentioned earlier.

2.2 Dashboards for data visualisation

The inclusion of a user-friendly dashboard is another critical component of our platform. This dashboard provides a comprehensive and intuitive display, presenting both the patient's visual experience within the VE and the signals collected through sensors on a single screen. Moreover, this interface facilitates various functionalities. It allows for recording therapy sessions, enabling clinicians to capture important moments for analysis and review. Additionally, clinicians can select specific time windows within the recorded video and add comments, either in real-time or offline. By incorporating this dashboard, we create a platform that encourages clinician-patient

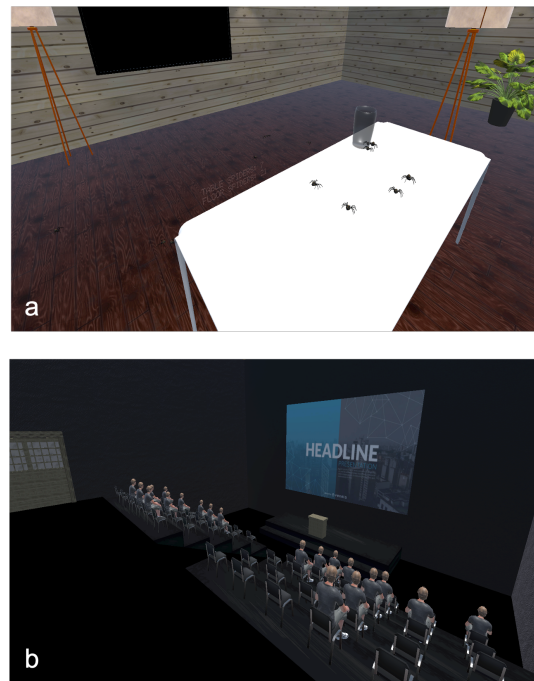


Figure 1: a) Arachnophobia scenario with spiders on the table; b) Public-speaking fear scenario, auditorium with audience.

Table 1: Main features identified for customisation in three VEs for VRET.

Type of scenario	Features
Arachnophobia	Type (room, bathroom, office) and environment's characteristics (light, doors) Type, size, and number of the spiders Animal behaviour characteristics (sound, speed, proximity, number) Tasks (e.g., turn off the light with a spider next to the light switch; trap the spider by putting a glass over it) Haptic feedback (e.g., controls vibrate when a spider is on patient's hand) Pre-emptive elements (signalling arrows, maps) Safe zones (e.g., spiders stay 2 meters away from the patients)
Fear of Public-Speaking	Type of environment (classroom, auditorium, congress hall) Number of people in the audience Audience facial expressions (e.g., serious vs smiling) and actions (e.g., laugh, gape, whisper, move in the chair or leave) Tasks (e.g., just stand in front of the audience vs given a presentation) Sound (complete silence, someone coughing, people talking)
Fear of Heights	Height and size of the platforms that patients have to walk through

dialogues regarding the most emotionally impacting moments experienced during therapy and facilitates a deeper understanding of the therapy process.

3 DTX CHALLENGES AND OPPORTUNITIES

Overall, the use of DTX worldwide is currently limited [2]. However, there is an anticipated significant increase in the adoption of DTX in the coming years. Two factors have propelled the integration of DTX into healthcare field. Firstly, unlike previous digital health apps, DTX are evidence-based software-driven therapeutic interventions [2]. This means that any DTX entering the market must provide claims of clinical efficacy supported by results obtained from randomised clinical trials (RCT). RCTs have served as the gold standard to establish the causality of an intervention (i.e., efficacy), as they minimise the risk of confounding factors and bias impacting the results [7]. Secondly, DTX are submitted to the same risk-based approach as other medical devices [13], ensuring compliance with strict specific indicators for safety, functionality, quality, data protection, data security, and clinical efficacy [6].

In order to stay abreast of the rapidly evolving technological field, regulatory agencies, such FDA (Food and Drug Administration) and DiGA (*Digitale Gesundheitsanwendungen*) have developed an innovative system for DTX certification, where what is assessed/certified are the development capabilities and safety of the development manufacturers rather than each individual medical device software [11].

By applying the methodologies employed in the assessment and regulation of drugs and medical devices to DTX, this new model can foster collaboration and understanding among data scientists, software developers, and clinicians. This approach promotes the



Figure 2: The control interface of the public-speaking fear scenario – a) number and localisation of people in the audience; b) audience movement; and c) emotional expressions conveyed through emojis.

use of a common language and shared objectives, bridging one of the gaps that has hindered the integration of DTX into clinical practice [2].

Furthermore, the approval of DTX by governmental institutions enables health professionals to prescribe these digital interventions, leading to coverage by health insurance and reimbursement by the Health Ministry. This broader accessibility ensures that a higher percentage of the general public can benefit from evidence-based digital interventions.

However, a challenge arises in the field of psychological interventions. Psychologists, unlike physicians who possess "prescription privileges" [1], may encounter limitations when it comes to

autonomously utilizing psychological DTx in their practice. This restriction poses obstacles in providing timely access to necessary interventions, contradicting the principle of ensuring wide accessibility to healthcare services through digital technologies [1]. Moreover, in extreme cases, the prolonged waiting period for patients to initiate their therapeutic program can potentially result in the deterioration of their mental state and exacerbation of existing barriers in the mental health field such as fear of stigma, reluctance to seek professional help, and early abandonment of the therapeutic intervention. In view of this, it is evident that a new approach to DTx regulation should be adopted to effectively address the diverse range of actions and advancements that psychological DTx will bring in the coming years.

4 CONCLUSION

In this paper, we present our vision on DTx for anxiety-related disorders in VRET. We describe a DTx that combines customising VEs and informative dashboards, aiming to enhance the effectiveness and efficiency of VRET in anxiety-related disorders. We believe this approach can potentially enhance VRET efficacy in anxiety treatment and improve patient outcomes. Additionally, we delve into the opportunities and threats associated with the application of a traditional medical prescription model to psychological DTx, and explore potential challenges that may arise as a result.

ACKNOWLEDGMENTS

This work was partially supported by FCT through LASIGE funding (ref. UIDB/00408/2020 and ref. UIDP/00408/2020) and Project 41, HfPT: Health from Portugal, funded by the Portuguese Plano de Recuperação e Resiliência.

REFERENCES

- [1] Jenna R Carl, Deborah J Jones, Oliver J Lindhiem, Brian D Doss, Kenneth R Weingardt, Adela C Timmons, and Jonathan S Comer. 2022. Regulating digital therapeutics for mental health: Opportunities, challenges, and the essential role of psychologists. *British Journal of Clinical Psychology* 61 (2022), 130–135.
- [2] Salvatore Crisafulli, Eugenio Santoro, Giuseppe Recchia, and Gianluca Trifirò. 2022. Digital therapeutics in perspective: from regulatory challenges to post-marketing surveillance. *Frontiers in Drug Safety and Regulation* 2 (2022), 900946.
- [3] Amit Dang, Dimple Arora, and Pawan Rane. 2020. Role of digital therapeutics and the changing future of healthcare. *Journal of Family Medicine and Primary Care* 9, 5 (2020), 2207.
- [4] Dhulika Dhingra and Aashima Dabas. 2020. Global strategy on digital health. *Indian pediatrics* 57 (2020), 356–358.
- [5] João Ferreira, Filipa Ferreira-Brito, João Guerreiro, and Tiago Guerreiro. 2022. Using VR and Sensors for Children and Adolescents. *Workshop on Virtual Reality for Health Wellbeing at 21thInternational Conf. Mob. Ubiquitous Multimed (MUM'22)* (2022).
- [6] Federal Institute for Drugs and Medical Devices. 2020. The Fast-Track Process for Digital Health Applications (DiGA) According to Section 139e SGB V.
- [7] Eduardo Hariton and Joseph J Locascio. 2018. Randomised controlled trials—the gold standard for effectiveness research. *BJOG: an international journal of obstetrics and gynaecology* 125, 13 (2018), 1716.
- [8] Kelly A Knowles and David F Tolin. 2022. Mechanisms of action in exposure therapy. *Current Psychiatry Reports* 24, 12 (2022), 861–869.
- [9] Katsunori Masaki, Hiroki Tateno, Akihiro Nomura, Tomoyasu Muto, Shin Suzuki, Kohta Satake, Eisuke Hida, and Koichi Fukunaga. 2020. A randomized controlled trial of a smoking cessation smartphone application with a carbon monoxide checker. *NPJ digital medicine* 3, 1 (2020), 35.
- [10] Mukherjee, S. 2017. <https://fortune.com/2017/09/14/fda-alcohol-marijuana-cocaine-mobile-app/>. Online; accessed June 2023.
- [11] Akihiro Nomura. 2023. Digital health, digital medicine, and digital therapeutics in cardiology: current evidence and future perspective in Japan. *Hypertension Research* (2023), 1–9.
- [12] Pietro Refolo, Dario Sacchini, Costanza Raimondi, and Antonio Gioacchino Spagnolo. 2022. Ethics of digital therapeutics (DTx). *European Review for Medical & Pharmacological Sciences* 26, 18 (2022).
- [13] U.S. Food & Drug Administration. 2022. <https://www.fda.gov/medical-devices/digital-health-center-excellence/device-software-functions-including-mobile-medical-applications>.